EXAM SOLUTIONS

1. a) True/False with justification.

i) **True.** By definition, for a real WSS process,

$$R_X(\tau) = E[X(t)X(t+\tau)] = E[X(t-\tau)X(t)] = R_X(-\tau),$$

where the second equality follows from the stationarity assumption (shift t by $-\tau$), and the third from commutativity of multiplication for real-valued processes. **[3 marks]**

ii) **True.** Linear operations preserve both Gaussianity and wide-sense stationarity. If the input to an LTI system is a WSS Gaussian process, the output is a linear functional (convolution) of the input, which remains Gaussian. Furthermore, the output of an LTI system driven by a WSS process is itself WSS (its mean is constant and its autocorrelation depends only on the time difference). **[3 marks]**

iii) **False.** Mutual information is always non-negative. This follows from the relation

$$I(X;Y) = H(X) - H(X|Y) \geq 0,$$

since conditioning cannot increase entropy: $H(X|Y) \leq H(X)$. Equality holds if and only if X and Y are independent. **[2 marks]**

iv) **False.** The capacity of an AWGN channel is given by $C = W\log_2\left(1+\frac{P}{N_0W}\right)$. Doubling W gives

$$C' = 2W\log_2\left(1+\frac{P}{2N_0W}\right).$$

Since $\log_2(1+x/2) < \log_2(1+x)$ for $x > 0$, we have $C' < 2C$. Intuitively, doubling the bandwidth also doubles the total noise power (for fixed N_0), so the SNR per unit bandwidth decreases, and capacity does not scale linearly with bandwidth. **[2 marks]**

b) Sampling and quantization of $m(t) = 3\cos(2\pi f_1 t) + 4\sin(2\pi f_2 t)$.

i) By the Nyquist sampling theorem, the minimum sampling rate for perfect recovery is $f_s = 2f_2 = 6$ kHz, since the highest frequency component is at $f_2 = 3$ kHz. **[2 marks]**

ii) The signal power is the sum of the powers of the individual sinusoidal components (they are at different frequencies, so there are no cross-terms):

$$P_S = \frac{3^2}{2} + \frac{4^2}{2} = \frac{9}{2} + \frac{16}{2} = \frac{25}{2} = 12.5.$$

[2 marks]

iii) The peak amplitude of $m(t)$ is at most $3+4=7$ (when both sinusoids add constructively). Hence the signal ranges over $[-7,7]$, and the peak-to-peak amplitude is 14. With 2^n quantization levels, the step size is

$$\Delta = \frac{14}{2^n}.$$

The quantization noise power for a uniform quantizer is

$$P_N = \frac{\Delta^2}{12} = \frac{196}{12 \cdot 4^n} = \frac{49}{3 \cdot 4^n}.$$

[3 marks]

iv) The signal-to-noise ratio is

$$\text{SNR} = \frac{P_S}{P_N} = \frac{25/2}{49/(3 \cdot 4^n)} = \frac{75 \cdot 4^n}{98}.$$

[2 marks]

v) We need $10\log_{10}(\text{SNR}) = 55.8$ dB, so $\text{SNR} = 10^{5.58} \approx 380{,}189$.

Setting $\frac{75 \cdot 4^n}{98} \geq 380{,}189$ gives $4^n \geq \frac{380{,}189 \times 98}{75} \approx 496{,}914$.

Then $2n \geq \log_2(496{,}914) \approx 18.92$, so $n \geq 9.46$. Hence the smallest integer value is $\boxed{n = 10}$. **[3 marks]**

c) White Gaussian noise through an ideal lowpass filter.

i) The ideal lowpass filter has frequency response $H(f) = 1$ for $|f| \leq B$ and $H(f) = 0$ otherwise. The output PSD is $S_Y(f) = |H(f)|^2 S_n(f)$, so

$$S_Y(f) = \begin{cases} \frac{N_0}{2}, & |f| \leq B, \\ 0, & |f| > B. \end{cases}$$

[2 marks]

ii) The autocorrelation function is the inverse Fourier transform of the PSD:

$$R_Y(\tau) = \int_{-B}^{B} \frac{N_0}{2} e^{j2\pi f\tau}\, df = N_0 B \,\mathrm{sinc}(2B\tau).$$

[3 marks]

iii) The total output noise power is

$$P_Y = R_Y(0) = N_0 B.$$

[2 marks]

d) DSB-SC modulation and demodulation.

i) The total power of the noise is

$$P_n = \int_{-\infty}^{\infty} S_n(f)\, df = 2\times 3\times 150 + 2\times 1\times 150 = 900 + 300 = 1200.$$

[2 marks]

ii) The signal power is $P_m = \int_{-W}^{W} S_m(f)\, df = W$ (the area of the triangular PSD).

For the DSB-SC receiver: after the bandpass filter centred at $f_c = 100$ with bandwidth $2W$, multiplication by $2\cos(2\pi f_c t)$, and lowpass filtering, the output signal power is $P_S = P_m = W$.

The output noise PSD (after demodulation) is $S_{n,\text{out}}(f) = S_n(f + f_c) + S_n(f - f_c)$ for $|f| \le W$.

For $W = 10$: the frequencies $f + f_c \in [90, 110]$ and $|f - f_c| \in [90, 110]$ both fall within $|f| \le 150$, where $S_n = 3$. Hence

$$S_{n,\text{out}}(f) = 3 + 3 = 6, \quad |f| \le 10.$$

The noise power is $P_N = 6 \times 2 \times 10 = 120$, and

$$\text{SNR} = \frac{P_S}{P_N} = \frac{10}{120} = \frac{1}{12}.$$

[3 marks]

iii) For $W = 80$: $P_S = 80$. The frequencies $f + f_c$ range over $[20, 180]$ as f ranges over $[-80, 80]$.

- For $|f| \le 50$: $f + f_c \in [50, 150]$ and $|f - f_c| \in [50, 150]$, so $S_n(f + f_c) = S_n(f - f_c) = 3$. Hence $S_{n,\text{out}}(f) = 6$.
- For $50 < |f| \le 80$: one of $f \pm f_c$ falls in the region $150 < |f| \le 300$ where $S_n = 1$, and the other in $|f| \le 150$ where $S_n = 3$. Hence $S_{n,\text{out}}(f) = 4$.

The noise power is

$$P_N = 6 \times 100 + 4 \times 60 = 600 + 240 = 840.$$

Therefore,

$$\text{SNR} = \frac{80}{840} = \frac{2}{21}.$$

[3 marks]

iv) With a phase offset of $\pi/4$ at the receiver, the demodulating carrier is $2\cos(2\pi f_c t + \pi/4)$. The product with the incoming DSB-SC signal $m(t)\cos(2\pi f_c t)$ gives:

$$2m(t)\cos(2\pi f_c t)\cos(2\pi f_c t + \pi/4) = m(t)\left[\cos(\pi/4) + \cos(4\pi f_c t + \pi/4)\right].$$

After lowpass filtering, only the baseband term survives: $y_s(t) = m(t)\cos(\pi/4) = \frac{m(t)}{\sqrt{2}}$.

The signal power is reduced to $P_S = P_m \cos^2(\pi/4) = 10 \times \frac{1}{2} = 5$.

The noise power is unchanged from part (ii): $P_N = 120$ (the noise analysis is not affected by the phase offset of the carrier).

Therefore,

$$\text{SNR} = \frac{5}{120} = \frac{1}{24}.$$

[3 marks]

2. a) WSS Gaussian process modulated by a random-phase carrier.

i) Since $X(t)$ is zero-mean and Θ is independent of $X(t)$,

$$E[Z(t)] = E[X(t)] \cdot E[\cos(2\pi f_0 t + \Theta)] = 0 \cdot E[\cos(2\pi f_0 t + \Theta)] = 0.$$

(Also, $E[\cos(2\pi f_0 t + \Theta)] = 0$ since Θ is uniform on $[0, 2\pi)$.)
[3 marks]

ii) We have

$$\begin{aligned} R_Z(t_1, t_2) &= E[Z(t_1)Z(t_2)] \\ &= E[X(t_1)X(t_2)] \cdot E[\cos(2\pi f_0 t_1 + \Theta)\cos(2\pi f_0 t_2 + \Theta)], \end{aligned}$$

where we used the independence of $X(t)$ and Θ. The first factor is $R_X(t_1 - t_2)$ by the WSS property of $X(t)$. For the second factor, we use the product-to-sum identity:

$$\begin{aligned} &E[\cos(2\pi f_0 t_1 + \Theta)\cos(2\pi f_0 t_2 + \Theta)] \\ &= \frac{1}{2}E[\cos(2\pi f_0 (t_1 - t_2))] + \frac{1}{2}E[\cos(2\pi f_0 (t_1 + t_2) + 2\Theta)] \\ &= \frac{1}{2}\cos(2\pi f_0 (t_1 - t_2)) + 0, \end{aligned}$$

since $E[\cos(2\pi f_0 (t_1 + t_2) + 2\Theta)] = 0$ when $\Theta \sim \text{Uniform}[0, 2\pi)$.

Letting $\tau = t_1 - t_2$, we get

$$\boxed{R_Z(\tau) = \frac{1}{2}R_X(\tau)\cos(2\pi f_0 \tau) = 2e^{-2|\tau|}\cos(2\pi f_0 \tau).}$$

Since $E[Z(t)] = 0$ (constant) and $R_Z(t_1, t_2)$ depends only on $\tau = t_1 - t_2$, the process $Z(t)$ **is WSS**. **[6 marks]**

iii) Using the modulation property of the Fourier transform, $S_Z(f) = \mathscr{F}\{R_Z(\tau)\}$:

$$S_Z(f) = \frac{1}{2} \cdot \frac{1}{2}[S_X(f - f_0) + S_X(f + f_0)] = \frac{1}{4}[S_X(f - f_0) + S_X(f + f_0)].$$

From the Fourier transform table, $R_X(\tau) = 4e^{-2|\tau|} \longleftrightarrow S_X(f) = \frac{16}{4 + (2\pi f)^2}$.

Hence

$$\boxed{S_Z(f) = \frac{4}{4 + 4\pi^2 (f - f_0)^2} + \frac{4}{4 + 4\pi^2 (f + f_0)^2}.}$$

[3 marks]

iv) The total power of $Z(t)$ is

$$P_Z = R_Z(0) = \frac{1}{2}R_X(0)\cos(0) = \frac{1}{2} \times 4 \times 1 = 2.$$

[3 marks]

b) BPSK matched filter detection.

i) The matched filter for the rectangular pulse $q(t)$ has impulse response $h(t) = q(T-t) = q(t)$. Hence the matched filter output at time T is

$$y(T) = \int_0^T r(t)\,q(t)\,dt = \int_0^T s_b(t)\,dt + \int_0^T n(t)\,dt.$$

The noise term $\int_0^T n(t)\,dt$ is Gaussian with zero mean and variance $\frac{N_0 T}{2}$. Therefore:

- For $b=1$: $y(T) = AT + \int_0^T n(t)\,dt$, so $y(T) \sim \mathcal{N}\left(AT,\ \frac{N_0 T}{2}\right)$.
- For $b=0$: $y(T) = -AT + \int_0^T n(t)\,dt$, so $y(T) \sim \mathcal{N}\left(-AT,\ \frac{N_0 T}{2}\right)$.

[3 marks]

ii) The bit error probability for threshold λ is

$$\begin{aligned} P_e &= P(b=0)\,P(y > \lambda \mid b=0) + P(b=1)\,P(y < \lambda \mid b=1) \\ &= P(b=0)\,Q\left(\frac{\lambda + AT}{\sqrt{N_0 T/2}}\right) + P(b=1)\,Q\left(\frac{AT - \lambda}{\sqrt{N_0 T/2}}\right). \end{aligned}$$

For equally likely bits ($P(b=0) = P(b=1) = 1/2$):

$$P_e = \frac{1}{2}Q\left(\frac{\lambda + AT}{\sqrt{N_0 T/2}}\right) + \frac{1}{2}Q\left(\frac{AT - \lambda}{\sqrt{N_0 T/2}}\right).$$

[4 marks]

iii) By symmetry of the problem (equal priors, equal signal energies, equal noise variances), the optimal threshold is $\lambda^* = 0$. This can also be verified by taking the derivative of P_e with respect to λ and setting it to zero. The resulting error probability is

$$P_e = Q\left(\frac{AT}{\sqrt{N_0 T/2}}\right) = Q\left(\sqrt{\frac{2A^2T}{N_0}}\right) = Q\left(\sqrt{\frac{2E_b}{N_0}}\right),$$

where $E_b = A^2 T$ is the energy per bit. **[4 marks]**

iv) Conditioned on α, the channel is a standard BPSK channel with amplitude αA, so the conditional error probability under threshold $\lambda^* = 0$ is

$$P_e(\alpha) = Q\left(\sqrt{\frac{2\alpha^2 E_b}{N_0}}\right).$$

Averaging over α:

$$\begin{aligned} \bar{P}_e &= \frac{3}{4}P_e(1) + \frac{1}{4}P_e\left(\tfrac{1}{2}\right) \\ &= \frac{3}{4}Q\left(\sqrt{\frac{2E_b}{N_0}}\right) + \frac{1}{4}Q\left(\sqrt{\frac{E_b}{2N_0}}\right). \end{aligned}$$

Since the Q-function is a decreasing function and $\sqrt{E_b/(2N_0)} < \sqrt{2E_b/N_0}$, we have $Q\left(\sqrt{E_b/(2N_0)}\right) > Q\left(\sqrt{2E_b/N_0}\right)$. This means the second term (corresponding to the faded case $\alpha = 1/2$) contributes disproportionately to the overall error probability.

Even though fading occurs only 25% of the time, it causes a substantial degradation: a factor-of-two attenuation in amplitude corresponds to a 6 dB loss in SNR. In general, fading degrades error performance because the increased errors during deep fades are not compensated by the (already low) error rate during good channel conditions, due to the nonlinear and rapidly decaying nature of the Q-function. **[4 marks]**

3. a) Source coding.

i) We apply the Huffman procedure. Sort by probability: $a(0.4)$, $b(0.3)$, $c(0.2)$, $d(0.1)$.

Step 1: Combine the two least probable symbols: $c(0.2)$ and $d(0.1) \to \{c,d\}(0.3)$.

New list: $a(0.4)$, $b(0.3)$, $\{c,d\}(0.3)$.

Step 2: Combine $b(0.3)$ and $\{c,d\}(0.3) \to \{b,c,d\}(0.6)$.

New list: $a(0.4)$, $\{b,c,d\}(0.6)$.

Step 3: Combine $a(0.4)$ and $\{b,c,d\}(0.6) \to \{a,b,c,d\}(1.0)$.

Assigning bits along the tree, one valid code is:

$$a \to 0, \quad b \to 10, \quad c \to 110, \quad d \to 111.$$

The average codeword length is

$$\bar{L} = 0.4\times 1 + 0.3\times 2 + 0.2\times 3 + 0.1\times 3 = 0.4+0.6+0.6+0.3 = 1.9 \text{ bits/symbol.}$$

The code is **not unique**: at Step 2, b and $\{c,d\}$ both have probability 0.3, so ties can be broken differently. However, all optimal prefix-free codes for this source achieve the same average length of 1.9 bits/symbol. **[4 marks]**

ii) Using $\log_2 3 = 1.58$ and $\log_2 5 = 2.32$:

$$\log_2(0.4) = \log_2(2/5) = 1 - 2.32 = -1.32$$

$$\log_2(0.3) = \log_2(3/10) = 1.58 - 1 - 2.32 = -1.74$$

$$\log_2(0.2) = \log_2(1/5) = -2.32$$

$$\log_2(0.1) = \log_2(1/10) = -1 - 2.32 = -3.32$$

$$\begin{aligned} H(S) &= -\sum_x P(x)\log_2 P(x) \\ &= 0.4\times 1.32 + 0.3\times 1.74 + 0.2\times 2.32 + 0.1\times 3.32 \\ &= 0.528 + 0.522 + 0.464 + 0.332 \\ &= 1.846 \text{ bits/symbol.} \end{aligned}$$

[3 marks]

iii) The coding efficiency is defined as the ratio of the source entropy to the average codeword length:

$$\eta = \frac{H(S)}{\bar{L}} = \frac{1.846}{1.9} \approx 97.2\%.$$

The coding efficiency can be improved by encoding *blocks* of source symbols (e.g., pairs or triples of consecutive symbols) and applying the Huffman procedure to these extended blocks. As the block length $k \to \infty$, the average number of bits per source symbol approaches the entropy $H(S)$, by Shannon's source coding theorem. **[3 marks]**

iv) Using the code $a \to 0, b \to 10, c \to 110, d \to 111$ for source T with $P(a) = 0.1$, $P(b) = 0.2$, $P(c) = 0.3$, $P(d) = 0.4$:

$$\bar{L}_T = 0.1 \times 1 + 0.2 \times 2 + 0.3 \times 3 + 0.4 \times 3 = 0.1 + 0.4 + 0.9 + 1.2 = 2.6 \text{ bits/symbol.}$$

Note that T has the same entropy as S (by symmetry, $H(T) = H(S) = 1.846$ bits/symbol), so the coding efficiency drops to $1.846/2.6 = 71.0\%$.

The code is **not optimal for** T: it assigns the shortest codeword (1 bit) to symbol a, which has the *lowest* probability under T. A Huffman code optimised for T would assign: $d \to 0, c \to 10, b \to 110, a \to 111$ (mirroring the original code), achieving $\bar{L} = 1.9$ bits/symbol. **[4 marks]**

b) Binary Erasure Channel.

i) For a given input $X = x$ (where $x \in \{0,1\}$), the output Y takes value x with probability $1-\varepsilon$ and value e with probability ε (and the other binary value with probability 0). Hence

$$H(Y|X=x) = -(1-\varepsilon)\log_2(1-\varepsilon) - \varepsilon \log_2 \varepsilon = H_b(\varepsilon),$$

where $H_b(\cdot)$ denotes the binary entropy function. Since this does not depend on x,

$$\boxed{H(Y|X) = p \cdot H_b(\varepsilon) + (1-p) \cdot H_b(\varepsilon) = H_b(\varepsilon).}$$

[3 marks]

ii) The output distribution is:

$$P(Y=0) = p(1-\varepsilon),$$
$$P(Y=e) = \varepsilon,$$
$$P(Y=1) = (1-p)(1-\varepsilon).$$

Therefore

$$H(Y) = -p(1-\varepsilon)\log_2[p(1-\varepsilon)] - \varepsilon \log_2 \varepsilon - (1-p)(1-\varepsilon)\log_2[(1-p)(1-\varepsilon)].$$

Expanding the logarithms:

$$H(Y) = -(1-\varepsilon)\big[p\log_2 p + (1-p)\log_2(1-p)\big] - (1-\varepsilon)\log_2(1-\varepsilon) - \varepsilon\log_2\varepsilon$$
$$= (1-\varepsilon)H_b(p) + H_b(\varepsilon).$$

[4 marks]

iii) The mutual information is

$$I(X;Y) = H(Y) - H(Y|X)$$
$$= \big[(1-\varepsilon)H_b(p) + H_b(\varepsilon)\big] - H_b(\varepsilon)$$
$$= (1-\varepsilon)H_b(p).$$

Since $H_b(p)$ is maximised at $p = 1/2$ (where $H_b(1/2) = 1$ bit), the channel capacity is

$$\boxed{C_{\text{BEC}} = \max_p I(X;Y) = 1-\varepsilon \text{ bits per channel use.}}$$

[5 marks]

iv) The capacity of the BSC with crossover probability ε is $C_{\text{BSC}}(\varepsilon) = 1 - H_b(\varepsilon)$.

We compare: $C_{\text{BEC}}(\varepsilon) = 1-\varepsilon$ versus $C_{\text{BSC}}(\varepsilon) = 1 - H_b(\varepsilon)$.

Since $H_b(\varepsilon) \geq \varepsilon$ for all $\varepsilon \in [0, 1/2]$ (the binary entropy function is concave and lies above the identity line), we have

$$C_{\text{BEC}}(\varepsilon) = 1-\varepsilon \;\geq\; 1 - H_b(\varepsilon) = C_{\text{BSC}}(\varepsilon),$$

with equality only at $\varepsilon = 0$ and $\varepsilon = 1$. The **BEC always has higher capacity**.

Intuitive explanation: In the BEC, when an erasure occurs the receiver *knows* the bit was lost — no misinformation is introduced. The receiver can simply discard erased bits and decode from the unerased ones. In the BSC, however, errors are "silent": the receiver gets a definite but *wrong* value and cannot tell which bits are corrupted. This undetected corruption is more harmful from an information-theoretic standpoint than a known loss, because it introduces *confusion* rather than merely reducing the amount of received data. **[4 marks]**